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EDITORIAL

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Artificial intelligence as digital agency

1. Wind of change

Over the last few years, the European Journal of Information Systems (EJIS) has gone through substantial changes (Ågerfalk, 2018). What, then, could be more appropriate than to set 2020 off on the same trajectory? 2018 saw the final year of Frantz Rowe's tenure as EJIS editor, and now the time has come to say goodbye also to Dov Te'eni. Words cannot describe how much I have enjoyed working with these two highly esteemed scholars – their contribution to EJIS and the information systems (IS) community[1] is impossible to overstate. Frantz was instrumental in orchestrating the scholarly mission of EJIS and establishing the current set of submission genres (Rowe, 2011, 2012). Both Frantz and Dov personify the diversity that characterises much of European IS research, which is also reflected by these genres (Te'eni, Seidel, & Vom Brocke, 2017). I trust their legacy will live on for a long time. Dov, I think, will be primarily remembered for two critical aspects of the kind of research that EJIS seeks to publish: contextualisation (Te'eni, 2015) and relevance to practice (Te'eni et al., 2017). Contextualisation can be understood as "the explicit addition of contextual information to a core message to ensure effective communication." (Katz & Te'eni, 2007, p. 261). By "considering and examining the context when planning and doing research" we can "arrive at a better understanding of the phenomena under question." (Te'eni, 2015, p. 362). The contextualisation of scholarly communication aimed at practitioners is key in order for IS to stay relevant to a practice audience (Te'eni et al., 2017), which is why EJIS, under Dov's leadership, spearheaded the Science to Practice Review initiative,[2] which is now being rolled out on a grander scale under the auspices of the Association for Information Systems.

No matter how sorry I am to see Dov leave, it is with equally great enthusiasm that I welcome Michael Myers on board as co-editor. Michael has until now served the journal as a senior editor with a particular emphasis on the ethnography/narrative genre. Michael is one of the intellectual giants of our field and I am truly honoured to have him by my side during my final year as EJIS editor-in-chief. His extensive experience as a scholar and editor will bring important lessons and new influences of which we expect to see ample evidence in the years to come. The future is looking bright for EJIS.

1.1. Te'eni's triad

Both Dov and Michael – I believe – share my view that IS inquiry is essentially about getting to grips with the design and use of information and communication technologies in practical contexts and critically uncovering the practices that emerge as digital technology is appropriated. The IS discipline is especially suited to such inquiry because of its balanced interest in digital technology, on the one hand, and social practice, on the other. This interest suggests that IS scholars should have a solid understanding of both digital technology and social practice, and especially of the "phenomena that emerge when the two interact." (Lee, 2001, p. iii).

Being contextual when studying digital practices is to black box neither technology nor practice. A key to achieving such transparency is to acknowledge that information systems are fundamentally digital mediators of human communication and symbolic action (Aakhus et al., 2014; Te'eni, 2001). In this view, IS inquiry emphasises three interrelated factors: contextualisation, communication, and practice. I refer to the combination of these three factors as Te'eni's Triad. Let us explore this triad and the concept of digital practice a bit further.

1.2. Digital practices

In an applied field such as IS, engaging with practice and studying practices comes rather naturally (Feldman & Pentland, 2003; Mathiassen, 2002; Orlikowski, 2000; Reckwitz, 2002; Sein, Henfridsson, Purao, Rossi, & Lindgren, 2011). A practice emerges when somebody uses resources to do something for someone in some context with some purpose. A practice is a digital practice when one or several of its constituents are digital. We can think of a digital somebody or someone (e.g. software agents, avatars, robots). One can use digital resources (e.g. digital data). Even the doing can be digital, by computational algorithms or by humans through digital media, or the doing may produce digital results. Even the context can, to a large extent, be digital (e.g. an agent-based system trading stocks), although such a context needs to be understood in relation to a social context with humans and intentions.

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As a result of the last few years of digitalisation and digital transformation, digital practices abound. We live in exciting times. Computing is everywhere, and digitalisation is changing almost every facet of life. The IS discipline should thrive. However, the steady stream of game-changing technologies seems often to be conceived of as a threat to the IS discipline rather than an opportunity – continuously fuelling the alleged IS identity crisis. The question becomes, "How should the IS discipline change in order to become relevant concerning this new technology?" rather than, "How can this technology be framed as relevant to IS inquiry?" Arguably, the most recent such technology spells Artificial Intelligence (AI).

1.3. The AI discourse

For a long time thought of as a computer science project with little relevance to IS, AI has come to take centre stage in the IS discourse as AI technologies are becoming increasingly practically relevant (Gomes et al., 2019). Indeed, it seems almost everybody is suddenly talking about AI (and becoming an AI researcher).

This is all good then? With AI left, right, and centre, IS should comfortably be leading the way towards new insight and improved practice, should it not? Or is AI becoming a management fad that has already lost its meaning? Is it not our responsibility as IS scholars to bring clarity to the discourse rather than contributing to its decline? To me, bringing clarity would mean not to talk about AI as a monolith, or even worse, just replacing IT or Technology with AI in our behavioural research models from yesteryear. It would mean to distinguish between different types of AI and not talk of AI as synonymous with machine learning, which in itself is far from a monolithic concept. The same goes for not confusing AI with analytics and big data, which are three different, although related beasts. It would also mean to avoid careless talk about "recent developments in AI" when referring to new applications of age-old AI techniques made practical by recent increases in processing power and storage capacity. After all, most current AI applications are based on machine learning that does pattern recognition. Animals can do pattern recognition.[3] In a cognitive sense, we seem to be pretty far from technological singularity (Makridakis, 2017) and AI as "the ability of a machine to perform cognitive functions that we associate with human minds" (Rai, Constantinides, & Sarker, 2019, p. iii). In a pragmatic sense, though, machines can already do things only humans used to do, and in some very specific tasks even outperform us.

1.4. Outline

In the following, I explore the concept of digital practice by addressing AI from an information systems perspective. The exploration serves a dual purpose. First, it addresses a potential role that IS can play in understanding AI and related technologies in social practices. Second, it acknowledges that the computational capability of digital technology calls for theorising that goes beyond sociological accounts of human agency. The exploration revisits some fundamental ideas about information technologies as mediators of symbolic action and communication. With this backdrop, the impact of AI on digital practices is framed as an IS problem of understanding digital agency, with consequences for how we may raise relevant IS research questions. The analysis should not be seen as a canonical attempt – nor as a proposal for a new IS nomological net (Benbasat & Zmud, 2003). My ambition is merely to show one way that IS can claim legitimacy in the AI discourse, and in so doing, also pay homage to Te'eni's Triad.

2. Information systems in context

To view AI and its related technologies from an IS perspective, it seems relevant to first establish what we mean by an information system. Although there are many different views on this topic (Alter, 2008; Avison & Elliot, 2006; Paul, 2010), a definition that I find resonates well with how we teach our subject to students is the one by Buckingham, Hirschheim, Land, and Tully (1987, p. 18), cited by Avison and Myers (1995): "a system which assembles, stores, processes, and delivers information relevant to an organisation (or to society), in such a way that the information is accessible and useful to those who wish to use it."

When studying information systems, one can consider a range of different aspects. From a material point of view, information systems are technological artefacts with physical properties (Goldkuhl & Ågerfalk, 2005; Orlikowski, 2007; Stamper, Liu, Hafkamp, & Ades, 2000). Because information systems are also semiotic artefacts that manipulate symbols and afford interpretation and communication (Aakhus et al., 2014; Stamper et al., 2000), they can be used as social instruments to express actors' intentions, values and beliefs (Goldkuhl & Ågerfalk, 2005; Hirschheim, Klein, & Lyytinen, 1996). As semiotic systems, information systems are also carriers of organisational institutions, enforcing and reinforcing norms, language and action (Aakhus et al., 2014; Stamper et al., 2000; Orlikowski & Scott, 2008). Through their semiotic symbol processing capability, they are also digital actors capable of performing social action on behalf of humans and organisations (Ågerfalk, 2004; Beynon-Davies, 2016). What this boils down to is that information systems need to handle not just decontextualised data but also socially meaningful signs. These signs are assembled, stored, transformed and delivered through our information

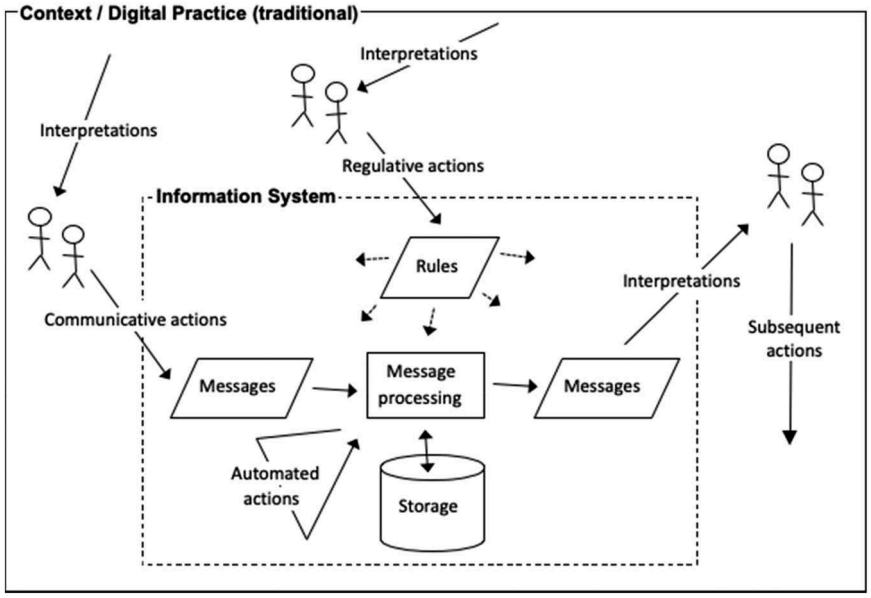


Figure 1. Information system in the traditional context. (After Goldkuhl & Ågerfalk, 2005).

systems. Goldkuhl and Ågerfalk (2005) summarised this view of an information system as an active mediator of social action and interaction as shown in Figure 1.

A critical aspect of this conceptualisation is the recognition of automated actions. Information systems perform these actions, which are comparable to the social actions performed by human actors. That is, they are actions in the sense of Max Weber's understanding of social action. Said Weber (1978, p. 4), "That action will be called 'social' which in its meaning as intended by the actor or actors, takes account of the behaviour of others and is thereby oriented in its course". An implication for information systems is that the signs, or messages, managed by these systems consist of more than a propositional content that describes the state of the world. The messages also embed intention and social meaning in terms of an illocutionary component (Ågerfalk, 2004; Ågerfalk & Eriksson, 2004) – they are essentially speech acts (Searle, 1969; Te'eni, 2006). Another implication for information systems is that these systems can perform social action with only indirect human intervention. Human actors govern the systems and their actions by rules stipulated through regulative actions. This way, information systems have agency in a way that goes beyond sociological notions of material agency, such as the notion of inscribed behaviour, widely used in the IS literature (Hanseth & Monteiro, 1997; Hultin & Mähring, 2014; Orlikowski & Scott, 2008).

With this understanding of an information system, IS inquiry is concerned with understanding the relationship between the social context in which people act and create meaning by formulating and interpreting messages, on the one hand, and the message processing performed by the system, on the other. An essential part of this processing is the automated actions that the system performs on behalf of its organisation. Thus, the delegation of actions from human actors, responsible for the social relationships created by those actions, and the system becomes a focal point (Ågerfalk, 2004). Although this conceptualisation of an information system was proposed well before the current surge of interest in AI, the delegation and autonomous performance of social actions are undoubtedly brought to the fore by AI. The question arises, how can this view of an information system bring clarity to the AI discourse? A key aspect of this, further discussed below, is to design for information systems accountability (Ågerfalk, 2004; Weitzner et al., 2008) – that is, to support actions that are "visibly-rational-and-reportable-for-all-practical-purposes" (Garfinkel, 1967, p. vii).

3. The contemporary information systems context

Understanding the relationship between AI and information systems requires an understanding of the contemporary information systems context. A first observation is that it is becoming increasingly difficult to talk about an individual information system without considering its connections to other information systems and digital resources of various kinds. This infrastructural character of information systems (Hanseth, 2010; Iannacci, 2010) is highlighted more than ever by internet of things (IoT) technology (Oberländer, Röglinger, Rosemann, & Kees, 2018) and digital platforms (de Reuver, Sørensen, & Basole, 2018). Figure 2 shows a high-level view of a digital

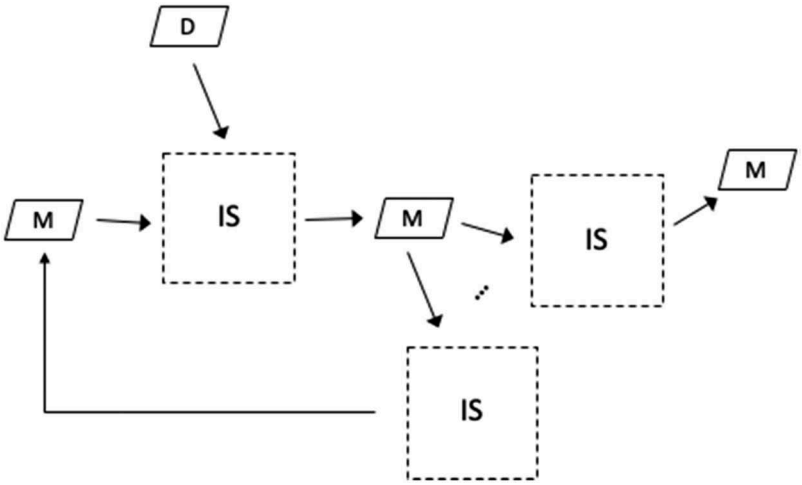


Figure 2. Interconnected information systems forming a digital information infrastructure. (D = data, M = message).

information infrastructure consisting of information systems, messages and data. As indicated above, data and messages are not the same thing. Data represent some empirical occurrence or phenomenon. Messages are symbolic constructs that embed data into socially meaningful units – messages carry intentions. Both data and messages can come from within the organisation as well as from external sources.

Returning to the information system of Figure 1 and appropriating it to the contemporary, infrastructural context of Figure 2, one may arrive at something like Figure 3. The differences between Figures 1 and 3 are to do with three things: emphasis on data, machine learning, and external sources and uses.

Data represent facts about empirical phenomena within or outside the current context. Big data discussions often focus on behavioural data collected about users of systems. The data we give off is typically separate from the messages we intend to communicate, but it can tell a lot about us and also help situate what we communicate (Ågerfalk & Sjöström, 2007). Said Goffman (1959, p. 2), "The expressiveness of the individual appears to involve two radically different kinds of sign activity: the expression that he gives, and the expression that he gives off". What we "give" are the messages we communicate. Our give-offs are the often non-verbal signs that help situate and verify the things we give – behavioural data, as it were. Another source of data that is becoming increasingly important

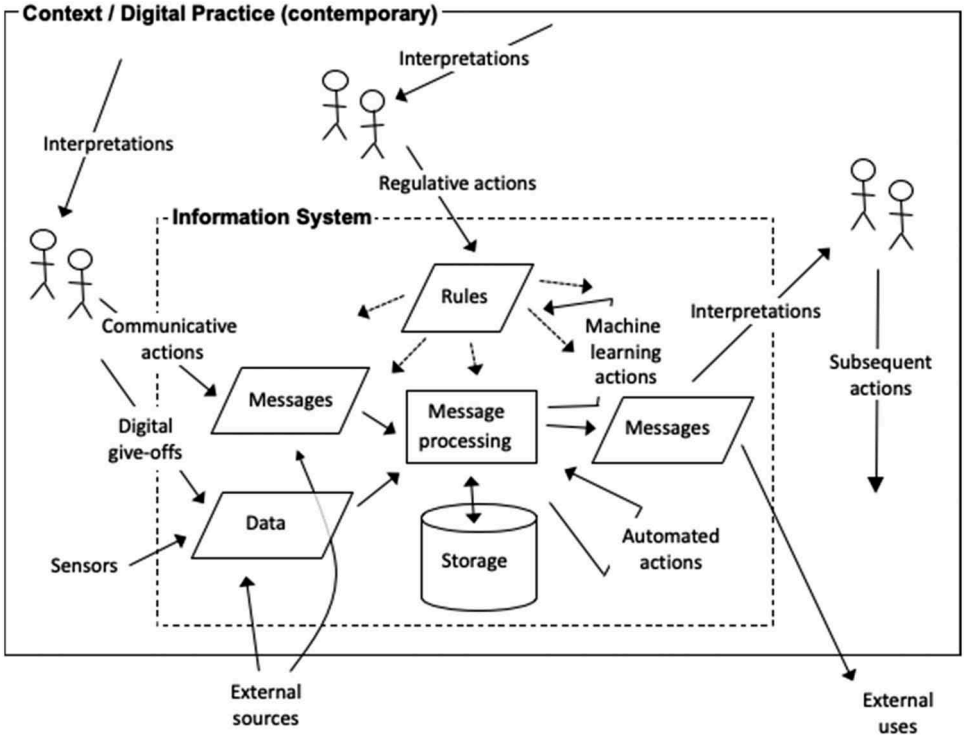


Figure 3. Information system in the contemporary context.

is different types of sensors, often part of networks of IoT devices that monitor various processes and events. The increasing use of big data for analytics (Abbasi et al. 2016; Sharma & Mithas, 2014) means that the information systems also need to consider decontextualised data and, through its message processing, attempt to make sense of the data for further processing and delivery of socially meaningful messages based on that data; i.e. effectively data mining.

What machine learning brings to the information systems picture is that these systems now can change their behaviour without direct human intervention. Essentially, machine learning is an automated regulative action. A machine learning action changes the rules under which the information system operates, just like how regulative actions by system developers stipulate the boundaries of what the system's message processing can do. This means that information systems contains rules that govern the stipulation of new rules, which in turn are governed by organisational rules and, ultimately, social and institutional norms and logics.

Contemporary information systems may source data and messages externally. External data may be any data that originates from outside the current practice and is input automatically, in real-time or in batches. External messages are part of structured message exchanges with external systems, such as EDI (Electronic Data Interchange) using ebXML. In the case of structured message exchange with an external system, the information system would usually also send messages externally to the interacting information system. External uses of messages could also be through APIs that external actors can use to get access to data from the information system, such as in open data applications.

Now, what does all this have to do with AI and Te'eni's Triad? Can this be a useful approach to bringing nuance to the IS discourse on AI? Maybe it could. The next section will explore some possible implications for IS inquiry.

4. Contextualising AI in IS

Te'eni's Triad reminds us not to forget the importance of contextualisation, communication, and practice. Contextualisation means that we should stay in touch with the practical context in which information systems are used. It also means that we should not black-box technology, and we should not assume that technologies will work the same or be ascribed the same meaning in all contexts. The message as a unit of analysis provides an instrument to connect data with its use in practical communication contexts. Data is meaningless unless connected to a social context. Messages are a way to make that connection explicit. Focusing on messages (as semiotic signs) and not just data, keeps focus on the contextual relevance of data and the uses of data in various practices. Messages are results of communicative actions. As such, they connect a communicator with some intended and actual interpreters (receivers). Behavioural data needs to be understood in relation to the social actions and the give-offs they represent. Data mining should not draw conclusions only based on patterns in the data without verifying that the illocution (social context) of that data is understood. A trivial example would be a data-mining application working on the assumption that everything that looks like an order is in fact an order. The data discovered may serve a completely different purpose, such as belonging to an invoice. A customer may communicate an order to the sales department. In contrast, an invoice may be communicated from the accounting department to the customer, or even to a third party in cases of invoice factoring. The practices in which these messages are created and interpreted are quite different. Certainly, some behavioural data can usefully be abstracted and attributed to the psychological characteristics of a fictive user. If we want to understand behavioural data pertaining to certain practices, however, the connection between message and give-off becomes critical. In any case, the transformation of data into socially meaningful messages is a delicate task that requires careful analysis of the overall communication context. IS people should be experts on this.

People talk about the "AI monster" as if it was something self-emerging and out of human control. Indeed, there are good reasons to worry about misuses of AI, big data, and large scale analytics, including "weapons of math destruction" (O'neil, 2016), "surveillance capitalism" (Zuboff, 2019), and the potential shaping of a new world order (Lee, 2018). These reasons are also why EJIS has launched a call for papers for a special issue on the dark side of analytics and AI.[4] However, an algorithm is an algorithm is an algorithm. Until we have reached technological singularity, humans develop algorithms (Makridakis, 2017). Machine learning as a form of automated action means that systems may modify their behaviour over time. The boundaries of such modifications are still managed by humans within technological, organisational and institutional frames. It is not a question of monsters but of agency concerning explainable and accountable AI. In social theory, agency traditionally defaults to human agency (Abdelnour, Hasselbladh, & Kallinikos, 2017), although some argue that behaviour can also be mediated by, and be inscribed in physical objects (Latour, 1991). Digital agency suggests something different: there is a difference between a dining table and Google's page ranking algorithm. The difference is that information systems are computational and therefore, can perform symbolic action. Agency is here seen as the capability of machines to act autonomously, but on behalf of humans, organisations and institutions. Digital agency does not presuppose consciousness in the traditional

sense. Instead, what becomes critical is the accountability of social actions performed by information systems. Computer scientists are working on explainable AI (XAI) as a way to un-black-boxing algorithmic decision making, especially concerning deep learning in layered neural network architectures. Developing XAI is necessary but not sufficient to address the digital-practice oriented questions related to accountable AI (Abdul, Vermeulen, Wang, Lim, & Kankanhalli, 2018). Who is responsible for the automatic actions performed? What social relationships are established by these actions, and why? On what grounds are actors performing subsequent actions based on messages from the system? What is the current and future role of human actors in formulating the regulative actions that govern machine learning actions? To what extent can we understand intentions and institutions by analysing decontextualised data? Answering these questions, and similar ones, requires an understanding of digital technology (such as XAI) and what XAI and other AI approaches mean to actors and organisations in the light of the institutional structures at work. These questions have immense practical implications and need urgent attention as practitioners and legislators are struggling to understand the consequences of AI (Ananny & Crawford, 2018; Diakopoulos, 2015) – an area where IS should be leading the way.

When discussing AI in relation to a conceptualisation of our primary object of study, the information system, I have indirectly suggested that what we see here are general problems that are only accentuated by AI. Information systems are computational systems. As such, they are active mediators of communication and social interaction, not passive media. Information systems are not merely information transmitters and data containers. This fundamental characteristic of digital technology calls for an ontological shift towards "digital first" (Aakhus et al., 2014; Baskerville, Myers, & Yoo, forthcoming; Eriksson, Johannesson, & Bergholtz, 2018). Information systems are not just representing an external reality; they are actively participating in and shaping reality (Beynon-Davies, 2016). Institutional facts (Searle, 2006) are increasingly being digitally created by automated actions defined by machine learning actions, which calls for rethinking institutionalisation altogether (Couldry & Hepp, 2017). Effectively, this ontological shift is at the core of the understanding of digital agency. IS scholars should seize the opportunity to theorise agency in a digital context without relying on outdated (pre-digital) notions of institutions and agency that do not pay attention to the computational capabilities of information systems. It seems we are witnessing a golden opportunity for IS truly to become a reference discipline (Baskerville & Myers, 2002).

At this point, I think Dov would probably object that I have not discussed how to bring these ideas into practice. To me, the answer is that to understand practice, one needs to engage with practice – a position that is part of our pragmatic outlook in IS (Ågerfalk, 2010). "In order for researchers to influence practice, they need to engage in dialog with practitioners. It is as simple as that." (Te'eni et al., 2017, p. 542). Thus, addressing the questions that emerge when looking at AI from an IS perspective naturally involves practice. I have previously argued that we need to acknowledge other types of contributions than theoretical ones (Ågerfalk, 2014), and AI in digital practices is no exception. An area where academic researchers will need practitioners, for instance, is in delineating contexts of investigation. Traditionally, the context would typically correspond to an organisation or department. The infrastructural perspective suggests that such intuitive borders are more challenging to define. We need multi-context studies where information systems, data, and messages are seen as boundary objects that connect contexts (Leigh Star, 2010). With digital agency and accountable AI on the table, studies may have to allow for contexts to continuously change and be redefined as these boundary objects become boundary agents. Somewhat ironically, industry is now leading the current "AI revolution" (Makridakis, 2017), despite the fact that AI concepts were originally developed by academics (Buchanan, 2005). IS researchers should engage in dialogue with these industrial AI practices and practitioners to help address questions such as those suggested above, and the ones yet to be asked.

5. Going out with a bang

I believe a disclaimer is in order. What I have presented in this editorial should not be seen as official EJIS policy but as representing this editor's current thinking about the role of IS in the age of AI. The EJIS editorial policy speaks of diversity and I encourage critical nitpicking of what I have presented and would expect and welcome future submissions to take completely different perspectives on the matter. I do hope, of course, that what I have presented can provide some food for thought in this emerging area. I furthermore suspect that neither Dov Te'eni nor Michael Myers would completely agree with my analysis. To be honest, Te'eni's Triad mainly served as a hook to create an excuse for me to expand on the role of IS in the AI discourse (and on the role of AI in the IS discourse). I do think it is a useful hook, though, and contextualisation, communication and practice are indeed three key components of IS inquiry in the European tradition.

Notes

1. I use the acronym IS when referring to the information systems discipline and the community of scholars belonging to that discipline. I spell out information systems when referring to the systems that these scholars study.
2. See https://science2practice.org/.

3. Bo Dahlbom during a keynote address at the 2019 annual meeting of the Swedish Information Systems Academy. Gothenburg, November 21 2019.
4. To read more about this special issue, including submission instructions, please visit https://www.tandfonline.com/ejis.

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